# CHUNG-YEH(OLIVER) YANG

(646) 992-7153 | cy2816@columbia.edu | linkedin.com/in/olivery0307 | https://github.com/Olivery0307

## EDUCATION

**Columbia University** New York, NY
**M.S. in Data Science,** GPA: 4.0/4.0 Dec 2026 (Expected)
Relevant Coursework: Agentic AI, Computer System, Computer Vision, Exploratory Data Analysis, Operational Logistics, Machine Learning, Poisson Process

**Boston University** Boston, MA
**B.S. in Data Science,** GPA: 3.8/4.0 Sep 2020 - Dec 2024
Major in Data Science, Minor in Business Administration
Coursework: Bayesian Statistics, Computer Systems, Database Design, Data Structure Algorithms, Deep Learning, Hypothesis Testing, Linear Algebra, Machine Learning, Multivariable Calculus

## WORK EXPERIENCE

**BCGX** New York, NY
Incoming Summer Forward Deployed AI Scientist June 2026 – August 2026
- Developed data and modeling components for a **network optimization** system for a national vehicle retailer, projecting a **15%** OTP improvement and **20%** cost reduction in backtesting
- Built a **Python/SQL** pipeline on **Snowflake** to clean and validate route, volume, and cost data, forming the team's core dataset
- Developed cost, time, and volume sub-models feeding a **Gurobi MILP** location optimizer for hub and lane routing decisions

**BU Spark!** Boston, MA
Data Science Technical Project Manager Sep 2024 - Dec 2024
- Managed six data analysis and visualization projects, coordinating with clients and facilitating student communication
- Provided technical assistance and leadership on data visualization projects leveraging **Python, Tableau,** and **Google BigQuery**
- Configured dataset and software setup for Notion and **GitHub**, ensuring teams had resources for project milestones

**GIGA RESET** Shanghai, China
Data Science Summer Intern Jun 2024 - Aug 2024
- Designed a data processing pipeline for the RESET Air project, which reduces manual workload by 80%
- Created Python scripts for visualizing air quality trends, identifying peaks, and generating reports for air quality
- Automated the creation of scorecards using Python, optimizing slide generation and data visualization

## PROJECTS

**Voyager – Travel Itinerary Optimizing Agent** Jan 2026– May 2026
- Architected a 7-agent travel planner (**OpenAI Agents SDK**) that converts NL prompt into bookable multi-city itineraries
- Designed **Pydantic**-typed tools across APIs and a replanner that re-optimizes on disruption while preserving locked bookings
- Validated on a golden-eval set spanning multi-city, weather-disrupted, and budget-constrained scenarios with 100% pass rate

**Text-to-SQL Data Intelligence Agent (Capstone Project)** Jan 2026 – May 2026
- Built a production-grade text-to-SQL chatbot using **LangGraph**, enabling business users to query databases conversationally
- Orchestrated a multi-agent pipeline, including intent router, SQL generation, and **ECharts** visualization for end-to-end analytics
- Engineered a self-evaluation retry-loop that lifted complex-query success from **70% to 100%** within three retry iterations

**YouBike Prediction Service** Jun 2025 – Apr 2026
- Developed a serverless data pipeline on **AWS (Lambda, S3)** to collect and process real-time YouBike API data
- Engineered time-series features to train a **LightGBM** model, achieving over 90% recall in predicting critical station states
- Deployed and dockerized and prediction pipeline using AWS Lambda and ECR for automated, live inference

## SKILLS

Programming Languages & Data Tools: Python, SQL, FastAPI, Apache Echarts, R, Redis, Tableau
AI/ML Frameworks: Pydantic, PyTorch, Multi-Agent Orchestration, OpenAI API, MCP/tool-calling, RAG, LangChain
Cloud & DevOps: AWS (Lambda, S3, ECR), Docker, Git, Google Cloud Platform (GCP), Supabase